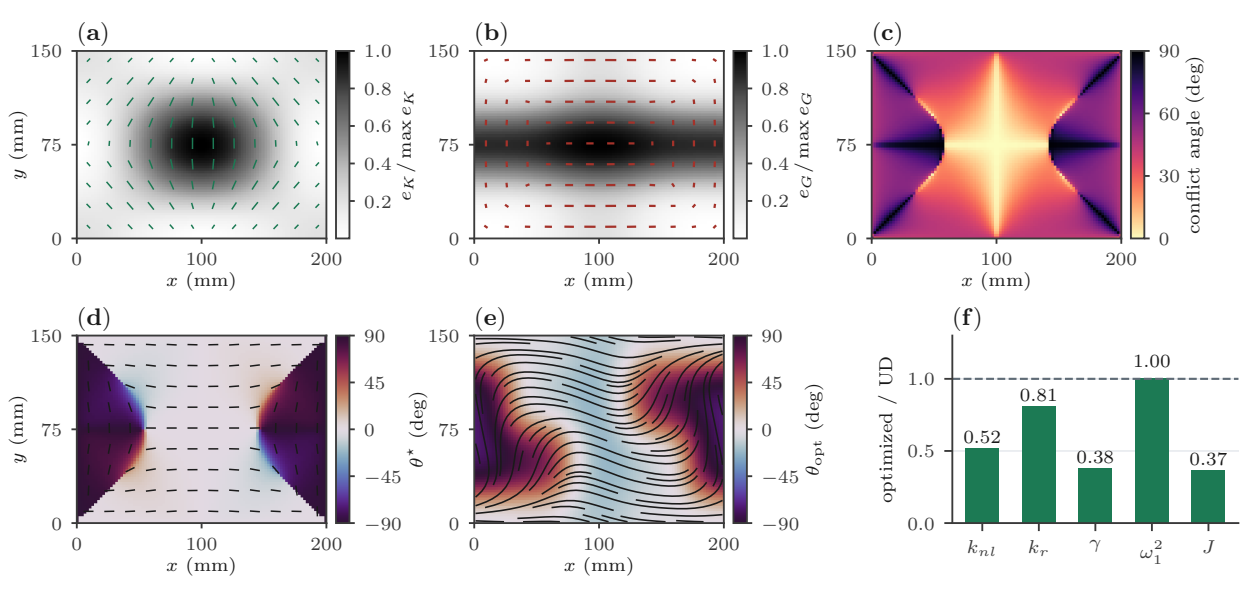

(a)
(b)
(c)
(d)
(e)
(f)
y (mm)
x (mm)
150
75
0
100
200
1.0
0.8
0.6
0.4
0.2
$e_K / \max e_K$
$e_G / \max e_G$
90
60
30
0
conflict angle (deg)
45
−45
−90
$\theta^\star$ (deg)
θ_{opt} (deg)
optimized / UD
0.5
0.52
0.81
0.38
1.00
0.37
k_{nl}
k_r
γ
ω_1^2
J